

2023



# Freecharge Transforms Its Digital Payment Platform on AWS to Improve Market Competitiveness

Freecharge modernized its application stack using Amazon DynamoDB and Amazon EKS, launching its consumer lending platform in 45 days and reducing the cost of disaster recovery by migrating to the AWS Asia Pacific (Hyderabad) Region.

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Hours to minutes

to create new environments



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# Overview

[Freecharge](#), a digital payments company wholly owned by Axis Bank, has been running its production environment on AWS since 2017. To maintain its edge in a highly competitive marketplace, Freecharge modernized its application stack on AWS.

Freecharge migrated its data store to [Amazon DynamoDB](#) and uses [Amazon EKS](#) to manage its Kubernetes clusters on AWS. The business also migrated its digital payments platform disaster recovery environments to the [AWS Asia Pacific \(Hyderabad\) Region](#), further optimizing costs and improving resiliency.



## Opportunity | Facilitating the Growth of Digital Payment Offerings

[Freecharge](#) is a digital payments company wholly owned by Axis Bank, the third largest private-sector bank in India. The company's payment app—Freecharge Wallet—is used by 100 million registered users across the country to send money and pay bills; Freecharge also offers a lending solution, a consumer credit card, merchant services, and an online payment gateway.

To maintain its competitive edge, Freecharge wanted to innovate its solutions, making it easier for consumers and businesses to use financial services. Rameshwar Gupta, chief technology officer at Freecharge, explains, "To stay competitive, we needed to get more out of our IT—lowering costs, going to market faster, and reducing administration where possible."

## Solution | Optimizing Architecture and Disaster Recovery with AWS

To improve the management of its Kubernetes clusters, Freecharge migrated its microservices environment to [Amazon Elastic Kubernetes Service](#) (Amazon EKS), achieving an increase in operational efficiency. These Kubernetes clusters house about 250 microservices across development, quality assurance, pre-production, and production environments, driving Freecharge's robust DevOps lifecycle.

"We achieved a 22 percent reduction in AWS costs and reduced deployment time from hours to just a few minutes, boosting team productivity," says Gupta. "This empowers Freecharge to accomplish multiple software releases daily, driving innovation and agility." Additionally, Freecharge decided to modernize its data store running on Aerospike, a NoSQL database engine that stored customer session data on [Amazon Elastic Compute Cloud](#) (Amazon EC2) instances. To solve issues relating to cost and scalability on Aerospike, Freecharge worked closely with AWS to migrate its data store to [Amazon DynamoDB](#), a secure, serverless [NoSQL database](#).

Vikalp Singh, director of infrastructure at Freecharge, says, "We seamlessly migrated from Aerospike to Amazon DynamoDB, achieving a 60 percent reduction in session data storage costs. This streamlined database scaling, freeing the business from the complexities of self-managed infrastructure."

Freecharge had previously used the AWS Asia Pacific (Mumbai) Region for its production environments since its launch in 2017. However, to comply with Reserve Bank of India (RBI) regulations mandating separate cities for primary and secondary sites, a third-party colocation site in Bangalore was employed for disaster recovery (DR). Upon the launch of the AWS Asia Pacific (Hyderabad) Region in March 2023, Freecharge migrated its disaster recovery (DR) environments for one of its critical applications.

Gupta says, "We reduced our DR costs by around 50 percent by moving out of our colocation site and into the AWS Asia Pacific (Hyderabad) Region, and by simplifying our DR processes." Furthermore, optimizing DR processes led to a 10–15 percent reduction in administrative time per DR drill, eliminating the need for complex configurations between AWS and the colocation site. "We can now efficiently conduct four routine DR drills annually without disrupting our operations," adds Gupta.

# Outcome | Driving Cost Optimization, Innovation, and Product Launches

Freecharge continues to work closely with AWS and [AWS Enterprise Support](#) to uncover new opportunities—channeling resources into expanding development teams, driving innovation, and accelerating product launches. The business launched its consumer lending platform in 45 days, a testament to the efficiency of Amazon EKS.

Through optimization, Freecharge also gained the time to develop its online payment gateways and cash advance solutions for business customers in a rapidly growing market. “We’re confident our IT infrastructure can keep pace with the business, even as we expect usage to increase 4–5 times over the next couple of years,” states Gupta. Furthermore, this support for expansion comes without a corresponding rise in IT expenditure.

“AWS Enterprise Support has been a tremendous help throughout our journey. AWS gets a perfect score from us for its support because it understands the challenges we face as a growing fintech business, particularly the need to continually manage costs,” concludes Gupta.

## Learn More

To learn more, visit [aws.amazon.com/products/databases/migrations](https://aws.amazon.com/products/databases/migrations).

## About Freecharge

Launched in 2010 and headquartered in Gurgaon, India fintech Freecharge provides app-based financial and investment services.

## AWS Services Used

### Amazon DynamoDB

Amazon DynamoDB is a fully managed, serverless, key-value NoSQL database designed to run high-performance applications at any scale.

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### Amazon Elastic Kubernetes Service



Amazon Elastic Kubernetes Service (Amazon EKS) is a managed Kubernetes service to run Kubernetes in the AWS cloud and on-premises data centers.

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## Amazon Elastic Compute Cloud

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

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## AWS Enterprise Support

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